

# Lecture V - Errors and Debugging

## Programming with Python

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### Checkpoint 2

**Before the doors open.** The first 40 minutes are the checkpoint. It starts now.

- **Individual work:** no AI, no neighbors, no chat
- The **link is on Moodle**. Open it and start
- ~6 short tasks: write code, trace code, fix a bug, answer a multiple choice
- It sweeps **Sessions III-IV**: functions, scope, classes, lists and dictionaries, comprehensions

### When you're done

Menu → *Download* → *Download Python code* → upload the `.py` to the “**Checkpoint 2**” assignment on Moodle, before the 40 minutes are up. **No retakes:** one sitting.

...

#### Note

The green live checks are **provisional**. The final grading runs on my side. Everything in it was rehearsed in the labs.

## Episode 5: The 3-AM Checkout

### After the checkpoint

Pens down. The checkpoint is over. On to Episode 5.

...

At 3 AM, running on his fourth energy drink, Tobi rewrote the **entire checkout** “to make it faster.” He went to bed very pleased. This morning two things are true: the till is a minefield of crashes, and the **health inspector** just called to announce a surprise visit.

...

So today the code learns to survive things going wrong. We read a **traceback**, catch failures with `try/except`, and let our own code refuse bad input.

# Reading the Explosion

## Tobi's 3-AM checkout, live

A customer typed `generous` into the tip box. Tobi's new checkout did this:

```
tip_text = "generous"
tip = float(tip_text) # crashes here
```

...

Python stopped and printed a **traceback**, a crash report. It tells you what went wrong and where:

```
Traceback (most recent call last):
  File "checkout.py", line 12, in <module>
    tip = float(tip_text)
ValueError: could not convert string to float: 'generous'
```

## Read the traceback bottom-up

You read a traceback from the **bottom up**:

- the **last line** names *what* went wrong (`ValueError`) and gives a hint
- the lines **above** show *where*: the file, the line number, the guilty code

|                                          |                                   |
|------------------------------------------|-----------------------------------|
| File "checkout.py", line 12, in <module> | ← WHERE it happened               |
| tip = float(tip_text)                    | ← the exact line                  |
| ValueError: could not convert ...        | ← WHAT went wrong (read me first) |

...

Two facts and you know where to go: **line 12**, and it's a `ValueError`. The rest of the traceback is context around those two facts.

## The big five (1)

A handful of exception types cover almost everything you'll hit. The first three:

- `ValueError`: right *type*, senseless *value*, as in `int("lots")`
- `TypeError`: the wrong type entirely, like `"Bowl " + 9`
- `KeyError`: a dictionary key that isn't there (`table_map["C3"]`)

...

Each one is Python refusing to guess. `int("lots")` has no sensible number; `"Bowl " + 9` mixes text and a number; `table_map["C3"]` asks for a key that was never added.

## Predict: comma or point?

A regular from Hamburg types the price the German way. What does the last line do?

```
price_text = "5,50"
price = float(price_text)
print(price)
```

a) prints 5.5 as a float b) raises `ValueError` c) raises `TypeError`

...

Predict first. Pick a letter, then I reveal the answer.

## Answer: right type, senseless value

b) raises `ValueError`: `float()` got what it expects, a string, so the *type* is fine. But "5,50" is not a number Python can read (it only knows the decimal point), so the *value* is refused. `TypeError` would need a non-string, like `float(None)`:

```
price_text = "5,50"
price = float(price_text)
print(price)
```

```
ValueError: could not convert string to float: '5,50'
```

```
[?] [31m-----[?]
[39m
[?] [31mValueError[?] [39m                                Traceback (most recent
call last)
[?] [36mCell[?] [39m[?] [36m [?] [39m[?] [32mIn[1][?] [39m[?] [32m, line 2[?] [39m
[?] [32m      1[?] [39m price_text = [?] [33m"[?] [39m[?] [33m5,50[?] [39m[?] [33m"[?] [39m
[?] [32m---->          [?] [39m[?] [32m2[?] [39m                price                =
[?] [38;5;28;43mfloat[?] [39;49m[?] [43m([?] [49m[?] [43mprice_text[?] [49m[?] [43m)[?] [49m
[?] [32m      3[?] [39m [?] [38;5;28mprint[?] [39m(price)

[?] [31mValueError[?] [39m: could not convert string to float: '5,50'
```

## The big five (2)

The other two you'll meet constantly:

- `IndexError`: a list position past the end, like `seats[9]` on a 3-item list
- `ZeroDivisionError`: dividing by zero, as in `88.00 / 0` when splitting a bill among no guests

...

You don't have to memorize all of Python's exceptions. Recognize these five on sight, and read the last line for the rest.

## Your turn: 5-10 minutes

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_05\\_a/](https://python.tobiasvlcek.com/notebooks/ex_05_a/)



First **predict** what happens, then run it.

### **try** / **except**: catch the fall

First, the happy path: text that **looks** like a number converts cleanly; text that doesn't, explodes. A **try** / **except** block runs the risky code and catches the failure instead of stopping the program:

```
print(float("5.50"))          # numeric text converts cleanly → 5.5

try:
    tip = float("generous")    # non-numeric text raises ValueError...
except ValueError:
    tip = 0.0                  # ...caught here, so the checkout keeps going

print(tip)
```

```
5.5
0.0
```

...

The risky line goes in the **try**; the recovery goes in the **except**. No crash. The program lands in the **except** and carries on.

## Catch the *specific* type

Name the exact exception you expect. `except ValueError:` catches **only** value errors and lets anything else through:

```
try:
    tip = float(tip_text)
except ValueError:          # only this kind
    tip = 0.0
```

...

### Warning

A **bare** `except:` catches *everything*, including your own typos (a misspelled variable would vanish silently). Catch the **specific** type, so real bugs still surface.

## Predict: which explosion?

Tobi builds a table label. `guests` is the number `3`. What does the **last line** do?

```
guests = 3
label = "Party of " + guests
print(label)
```

a) raises `TypeError` b) prints `Party of 3` c) raises `ValueError`

...

Predict first. Pick a letter, then I reveal the answer.

## Answer: text and numbers won't mix

a) raises `TypeError: +` can glue two strings or add two numbers, but it refuses to mix a string and an `int`. The fix is to convert first (`str(guests)`):

```
guests = 3
try:
    label = "Party of " + guests
except TypeError as e:
    print("TypeError:", e)
```

```
TypeError: can only concatenate str (not "int") to str
```

...

`as e` keeps the error object in a variable, so you can print its message instead of losing it. Any name works; `e` is the habit.

## Your turn: 5-10 minutes

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_05\\_b/](https://python.tobiasvlcek.com/notebooks/ex_05_b/)



First **predict** what happens, then run it.

## Debugging and Defending

### The debugging loop

When something is wrong, you don't guess randomly. You follow a loop:

1. **Read** the traceback: it hands you the *what* and the *where*
2. **Reproduce** the crash: make it happen on demand
3. **Isolate**: narrow down to the smallest failing piece
4. **Fix**: change one thing, then run again

...

The inspector debugs the same way: find the broken thing, make it repeat, corner it, fix it. Panic is not a step.

### `print()`: the beginner's flashlight

The simplest debugger is a well-placed `print()`. When you can't see what a value *is*, shine a light on it:

```
def line_total(qty, price):
    print("DEBUG qty =", qty, "price =", price)    # the flashlight
    return round(qty * price, 2)

print(line_total(3, 5.50))
```

```
DEBUG qty = 3 price = 5.5
16.5
```

...

Real debuggers with breakpoints exist and are wonderful, but a `print()` in the right spot solves most beginner bugs in seconds. Remove it once you've seen enough.

### **raise: when your code should refuse**

Sometimes *your own* code should reject bad input on the spot. The inspector's rule is non-negotiable: **every price  $\geq 0$** . `raise` throws an error deliberately:

```
def charge(amount):
    if amount < 0:
        raise ValueError("price cannot be negative")
    return amount

try:
    charge(-4.50)                # your code refuses it...
except ValueError as e:
    print("refused:", e)         # ...on your terms
```

```
refused: price cannot be negative
```

...

A bad value stops *here*, at the door, instead of poisoning the books three screens later.

### **assert: a tripwire for invariants**

An `assert` guards an *invariant*, a fact that must **always** hold. If it's true, nothing happens; if it's false, the program stops right there with an `AssertionError`:

```
subtotal = 9.60
assert subtotal >= 0, "subtotal went negative: bug upstream"
print("passed the tripwire:", subtotal)
```

```
passed the tripwire: 9.6
```

...

Think of it as a note to yourself, checked automatically: *"if this is ever false, something broke earlier. Stop before it spreads."*

### **Predict: does the `print` survive?**

The bill is being split, but nobody is at the table. What shows up?

```

guests = 0
assert guests > 0, "no guests at the table"
print("splitting the bill among", guests)

```

a) `AssertionError`, and the `print` never runs b) prints `splitting the bill among 0` c) the message, then the `print` line as well

...

Predict first. Pick a letter, then I reveal the answer.

## Answer: the tripwire stops everything

a) `AssertionError`, and the `print` never runs: a failing `assert` raises on the spot, and nothing after it in the program executes. That's the point of a tripwire: stop *before* a wrong number spreads:

```

guests = 0
assert guests > 0, "no guests at the table"
print("splitting the bill among", guests)

```

```

AssertionError: no guests at the table

```

```

[31m-----[?]
[39m
[31mAssertionError[39m                                Traceback (most recent
call last)
[36mCell[39m[36m [39m[32mIn[7][39m[32m, line 2[39m
[32m      1[39m guests = [32m0[39m
[32m----> [39m[32m2[39m [38;5;28;01massert[39;00m guests >
[32m0[39m, [33m"[39m[33mno guests at the table[39m[33m"[39m
[32m      3[39m [38;5;28mprint[39m([33m"[39m[33msplitting the
bill among[39m[33m"[39m, guests)

[31mAssertionError[39m: no guests at the table

```

## Try, then fall back

A friendly, common pattern: **try** the risky thing; if it fails, **fall back** to a sensible default so the program keeps moving.

```

def seats(text):
    try:
        return int(text)
    except ValueError:
        return 2                # a table seats two, unless told otherwise

```

```
print(seats("6"))           # a clean number → 6
print(seats("full"))        # nonsense → the safe default, 2
```

```
6
2
```

...

The caller never has to worry about a crash. Garbage in, sane default out. Not every check needs a `try: isinstance(x, (int, float))` asks *is this a number?* and returns `True/False`, so an `if` can skip bad values before they explode.

### Predict: what happens after the `except`?

`to_price` catches a bad value and returns `0.0`. After that, does the program reach the next line?

```
def to_price(text):
    try:
        return float(text)
    except ValueError:
        return 0.0

print(to_price("bad"))
print("checkout still running")
```

a) it never runs: the program already stopped b) the `try` block runs a second time c) it runs normally: the program carried on

...

Predict first. Pick a letter, then I reveal the answer.

### Answer: it carries on

**c) it runs normally.** That's the whole point of `try/except`. It *handles* the failure; once the `except` has run, control simply drops to the code after it, as if nothing had gone wrong:

```
def to_price(text):
    try:
        return float(text)
    except ValueError:
        return 0.0

print(to_price("bad"))           # 0.0, recovered
print("checkout still running") # ...and we get here
```

```
0.0
checkout still running
```

## Your turn: 5-10 minutes

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_05\\_c/](https://python.tobiasvlcek.com/notebooks/ex_05_c/)



First **predict** what happens, then run it.

## To the Lab

### After the break: the lab

- Head to the lab notebook: [Episode 5: The 3-AM Checkout](#)
- You'll read a traceback, write a price box that won't crash, fix Tobi's 3-AM receipt, make your code **refuse** negative prices with `raise`, and harden the whole checkout against a batch of poisoned orders
- It runs entirely in your browser: no setup, just click and code

...

#### ! Important

**Download your `.py` before you leave.** Closing the tab without downloading loses your work, and downloading is exactly how you just handed in the checkpoint. Expect **red cells** in this lab: you'll cause errors on purpose. A red cell pauses everything below it. Fix it, and everything springs back.

## Wrap-up

### Three things to remember

1. A **traceback** is a crash report you read **bottom-up**: the last line names *what* broke, the lines above show *where*
2. **try / except** catches a failure so the program recovers instead of stopping. Catch the specific type (`except ValueError`), never a bare `except` that also hides your own typos
3. Defend your own code: **raise** to refuse bad input (prices  $\geq 0$ ), **assert** to guard an invariant that must always hold, and fall back to a safe default when a conversion fails

...

#### Note

**Next time, Episode 6** starts with Checkpoint 3, which sweeps everything from Episodes 1-5, so keep this notebook and the last four close. Then someone new walks into the shop, uncaps a marker, and writes one question on the whiteboard. Part II begins.

## Literature

### Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

...

#### Note

Nothing new here, but these are still great books to start with!

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For more, see the [literature list](#) of this course.